CS604 Operating System
Current Paper Spring 2021
Final-Term
SUBJECTIVE

If we take an average page fault service time of 25 milliseconds and a memory access time of 100 nanoseconds, then the effective access time?

Ans:

Effective access time = (1-p) * (100) + p (25 milliseconds)
= (1-p) * 100 + p * 25,000,000
= 100 + 24,999,900 * p

If HR is hit ratio and MR is miss ratio, the effective access time is given by the following equation Teffective = HR (TTLB + Tmem) + MR (TTLB + 2Tmem)

Ans:

Tmem = 100 nsec
TTLB = 20 nsec
Hit ratio is 80%
Teffective = 0.8 (20 + 100) + 0.2 (20 + 2*100) nanoseconds = 140 nanoseconds

Write the formula to calculate page table size?

Ans:

Page table size = NP * PTES

Perform semaphore using P0 and P1 for starvation.

Ans:

P0
wait(S);
…
wait(S);
…

P1
wait(S);

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....
single(S);
...

Assume a logical address space of 16 pages of 1024 words, each mapped into a physical memory of 32 frames. Here is how you calculate the various parameters related to paging? Find Logical address and Physical address size?

Ans:

No. of bits needed for p = ceiling [log2 16] bits = 4 bits
No. of bits needed for f = ceiling [log2 32] bits = 5 bits
No. of bits needed for d = ceiling [log2 2048] bits = 11 bits
Logical address size = |p| + |d| = 4+11 bits = 15 bits
Physical address size = |f| + |d| = 5+11 bits = 16 bits

Does following producer's code have any ambiguity and correct?

Ans:

do {
...
produce an item in nextp
...
wait(empty);
wait(mutex);
...
add nextp to buffer
...
signal(mutex);
signal(full);
} while(1);

How can we calculate a size of page table? Just provide the formula?

Ans:

Page table size = NP * PTES, where NP is the number of pages in the process address space and PTES is the page table entry size.

Define lazy swapper with respect to virtual memory?

Ans:

A lazy swapper never swaps a page into memory unless that page will be needed.

How page fault frequency model used to control the thrashing?

Ans:

Page fault frequency is another method to control thrashing. Since thrashing has a high page fault rate, we want to control the page fault frequency. When it is too high we know that the process needs more frames. Similarly if the page-fault rate is too low, then the process may have too many frames. The operating system keeps track of the upper and lower bounds on the page-fault rates of processes. If the page-fault rate falls below the lower limit, the process loses frames. If page-fault rate goes above the upper limit, process gains frames. Thus we directly measure and control the page fault rate to prevent thrashing.

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What is difference between deadlock avoidance and deadlock prevention?

Ans:

Deadlock prevention: is a set of methods for ensuring that at least one of the necessary conditions cannot hold. These methods prevent deadlocks by constraining how processes can request for resources.

Deadlock Avoidance: This method of handling deadlocks requires that processes give advance additional information concerning which resources they will request and use during their lifetimes. With this information, it may be decided whether a process should wait or not.

What is swap space?

Ans:

The secondary memory is usually a high speed disk. It is known as the swap device, and the section of disk used for this purpose is called the swap space.

Drawbacks of Semaphore?

Ans:

- Simple algorithms require more than one semaphore
- This increases the complexity of semaphore solutions to such algorithm
- The programmer must keep track of all calls to wait and to signal the semaphore.
- Since semaphores can be tricky, can we create other constructs from them

Semaphores Algorithm?

Ans:

A semaphore S is an integer variable that, apart from initialization is accessible only through two standard atomic operations: wait and signal. These operations were originally termed P (for wait) and V (for signal). We can use semaphores to deal with the n-process critical section problem. The n processes share a semaphore, mutex (standing for mutual exclusion) initialized to 1.

Busy Waiting?

Ans:

The main disadvantage of the semaphore discussed in the previous section is that they all require busy waiting. Busy waiting wastes CPU cycles that some other process may be able to use productively. This type of semaphore is also called a spinlock (because the process spins while waiting for the lock).

The advantage of a spinlock is that no context switch is required when a process must wait on a lock, and a context switch may take considerable time. This, when locks are expected to be held for short times, spinlocks are useful.

Consider a logical address space of eight pages of 1024 words each, mamapped ont a Physical memory of 32 frames.

(a)How many bits are there in the logical address?

(b)How many bits are there in the physical address?

Ans:

Each page/frame holds 1K; we will need 10 bits to uniquely address each of those 1024 addresses. Physical memory has 32 frames and we need 2 5 bits to address each frames, requiring in total 5+10=15 bits. A logical address space of 64 pages requires 6 bits to address each page uniquely, requiring 16bits in total.

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Reference: http://garryowen.csisdmz.ul.ie/~cs4023/resources/sol11.pdf

Deadlock Detection?

Ans:

If a system does not employ either a deadlock prevention or a deadlock avoidance algorithm then a deadlock may occur. In this environment, the system must provide:

- An algorithm that examines (perhaps periodically or after certain events) the state of the system to determine whether a deadlock has occurred
- A scheme to recover from deadlocks.

Overlays problems?

Ans:

The problems with overlays are that

- you may not be able to partition all problems into overlays, and
- Programmer is responsible of writing the overlays driver.

What is Index Allocation Method?

Ans:

Indexed allocation brings all the pointers to the block together into a disk block, known as the index block. Indexed allocation supports direct access without suffering from external fragmentation because any free block on the disk may satisfy a request for more space.

Who generate physical address and logical address?

Ans:

An address generated by the CPU is commonly referred to as a logical address, where as an address seen by the memory unit–that is, the one loaded into the memory-address register of the memory–is commonly referred to as the physical address. In essence, logical data refers to an instruction or data in the process address space where as the physical address refers to a main memory location where instruction or data resides. The compile time and load time binding methods generate identical logical and physical addresses, whereas the execution time binding method results in different physical and logical addresses. In this case we refer to the logical address as the virtual address. The set of all logical addresses generated by a program form the logical address space of a process; the set of all physical addresses corresponding to these logical addresses is a physical address space of the process. The total size of physical address space in a system is equal to the size of its main memory. The run-time mapping from virtual to physical addresses is done.

4 Ways for file sharing?

Ans:

- Duplicating files
- Common login
- Access permissions
- Common groups
- Links

How we do memory protection in memory management?

Ans:

Using two CPU registers, specifically designed for this purpose, can provide memory protection. These registered are:

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Base register – it holds the smallest legal physical memory address for a process
Limit register – it contains the size of the process.

Is starvation and deadlock are same. Accept or reject with solid reason?

Ans:

No! Two neighbors are eating simultaneously; it nevertheless must be rejected because it has the possibility of creating a deadlock. A set of processes are said to be in a deadlock state if every process is waiting for an event that can be caused only by another process in the set and Starvation is infinite blocking caused due to unavailability of resources.

Deadlock handling?

Ans:

In an operating system, a deadlock occurs when a process or thread enters a waiting state because a requested system resource is held by another waiting process, which in turn is waiting for another resource held by another waiting process.

Deadlock handling and avoidance

Deadlock prevention. Prevents deadlocks by restraining requests made to ensure that at least one of the four deadlock conditions cannot occur. **Deadlock avoidance**. Dynamically grants a resource to a process if the resulting state is safe.

Deadlock detection?

Ans:

Deadlock detection is the process of actually determining that a deadlock exists and identifying the processes and resources involved in the deadlock. ... Once a deadlock is detected, there needs to be a way to recover several alternatives exists: Temporarily prevent resources from deadlocked processes.

Preemptive and non-preemptive algorithms?

Ans:

Preemptive scheduling allows a running process to be interrupted by a high priority process, whereas in non-preemptive scheduling, any new process has to wait until the running process finishes its CPU cycle. We have also discussed the differences between preemptive and non-preemptive scheduling.

Three types of access modes and classes of users in UNIX protection?

Ans:

UNIX recognizes three modes of access: read, write, and execute (r, w, x). The execute permission on a directory specifies permission to search the directory

The three classes of users are:

- **Owner**: user is the owner of the file
- **Group**: someone who belongs to the same group as the owner ϖ
- **Others**: everyone else who has an account on the system.

Write the steps for converting source code to executable form?

Ans:

Translation of a source program in a high-level or assembly language involves compilation and linking of the program series of steps:

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Compile/Assemble

↓

Link

↓

Load

↓

Execute

Name three access modes use in file protection?

Ans:

- Read
- Write
- Execute

Basic logic in FIFO page replacement algorithm?

Ans:

The simplest page-replacement algorithm is a FIFO algorithm. A FIFO replacement algorithm associates with each page the time when that page was brought into memory. When a page must be replaced, the oldest page is chosen. Notice that it is not strictly necessary to record the time when a page is brought in. We can create a FIFO queue to hold all pages in memory. We replace the page at the head of the queue. When a page is brought into memory we insert t at the tail of the queue.

Paged Segmentation?

Ans:

In paged segmentation, we divide every segment in a process into fixed size pages. We need to maintain a page table per segment CPU's memory management unit must support both segmentation and paging.

What is Scheduling?

Ans:

The process scheduling is the activity of the process manager that handles the removal of the running process from the CPU and the selection of another process on the basis of a particular strategy. Process scheduling is an essential part of a Multiprogramming operating systems.

Earned Value Analysis?

Ans:

Earned Value Analysis (EVA) is a method that allows the project manager to measure the amount of work actually performed on a project beyond the basic review of cost and schedule reports. EVA provides a method that permits the project to be measured by progress achieved.

Write the names of common file structures?

Ans:

- None
- Simple record structure
- Complex Structures

CPU sends the user or application request to the kernel. How kernal works on the request and fulfill it?

Ans:

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Any user or application request that involves access to any system resource must be handled by the kernel code, but user process must not be given open access to the kernel code for security reasons.

Define mounting in Unix?

Ans:

All files accessible in a Unix system are arranged in one big tree, the file hierarchy, rooted at /. These files can be spread out over several devices. The mount command serves to attach the file system found on some device to the big file tree.

Sometimes mkfifo call may be failure, write the reasons for failure if mkfifo call in Unix?

Ans:

- File with the given name already exists
- Pathname too long
- A component in the pathname not searchable
- Non-existent or non-directory
- Destination directory is read-only
- Not enough memory space available
- Signal caught during the execution of mkfifo()

What does the following command do in the LINX/UNIX operating system $mkdir ~/courses/cs604/program?

Ans:

Command creates the programs directory under your ~/courses/cs604 directory.

List down 2 major benefits of virtual memory?

Ans:

- Virtual Memory is the separation of user logical memory from physical memory. This separation allows an extremely large virtual memory to be provided for programmers when only a smaller physical memory is available.
- Virtual memory makes the task of programming easier because the programmer need not worry about the amount of physical memory.

How to detect cycles in acyclic graph?

Ans:

A solution is to allow only links to files not subdirectories. Also every time a new link is added use a cycle detection algorithm to determine whether it is OK. If cycles are allowed, we want to avoid searching any component twice.

What is lazy analyzer that is use in virtual memory?

Ans:

When we want to execute a process, we swap it into memory. Rather than swapping the entire process into memory, however we use a lazy swapper.

Name of two registers used in segmentation?

Ans:

- Segment-table base register (STBR)
- Segment-table length register (STLR)

Does cooperating processes are helpful to the operating system?

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Ans:

The concurrent processes executing in the operating system may be either independent processes or cooperating processes. A process is independent if it cannot affect or be affected by any other process executing in the system. Clearly any process that shares data with other processes is a cooperating process.

Give one advantage and one disadvantage of using large sized block?

Ans:

Advantage

- Good for large rows with lots of sequential I/O
- Has a lower overhead with less block headers so more room to store data
- Well suited for tables with large sized rows

Disadvantage

- Wastes space in the buffer cache if doing random I/O to access a small number of rows
- In OLTP systems can increase block contention due to more rows being stored in each block

If we create a pipe and a pipe return a numeric value -1 what does it means?

Ans:

The function returns -1 if the call fails.

Differentiate between logical and physical address?

Ans:

1. The basic difference between Logical and physical address is that Logical address is generated by CPU in perspective of a program whereas the physical address is a location that exists in the memory unit.
2. Logical Address Space is the set of all logical addresses generated by CPU for a program whereas the set of all physical address mapped to corresponding logical addresses is called Physical Address Space.
3. The logical address does not exist physically in the memory whereas physical address is a location in the memory that can be accessed physically.
4. Identical logical addresses are generated by Compile-time and Load time address binding methods whereas they differs from each other in run-time address binding method. Please refer this for details.
5. The logical address is generated by the CPU while the program is running whereas the physical address is computed by the Memory Management Unit (MMU).

A code was given and question was to tell whether it satisfies mutual exclusion?

Ans:

An algorithm solves the mutual exclusion problem if the following hold:

Mutual Exclusion: In every configuration of every execution, at most one process is in the critical section.

No Deadlock: In every execution, if some process is in the entry section in some configuration, then there is a later configuration in which some process is in the critical section.

Stronger Progress Property.

No lockout (starvation-free): In every execution, if some processor is in the entry section in a configuration, then there is a later configuration in which that same processor is in the critical section.

Reference: http://www.csd.uoc.gr/~hy586/material/lectures/cs586-Section2.pdf

Soft links in UNIX?

Ans:

Soft links take care of all the problems inherent in hard links. They are flexible. You may have soft links to directories and across file systems. However, UNIX has to support an additional file type, the link type, and a new file is created for every link, slowing down file operations.

How can you calculate size of page table, give formula?

Ans:

Page table size = NP * PTES , where NP is the number of pages in the process address space and PTES is the page table entry size (equal to |f| based on our discussion so far). Page table size = 16 * 5 bits

Which factors are determine to choose a process for Termination?

Ans:

Many factors determine which process is chosen, including:

7) What the priority of the process is

8) How long the process has computed, and how much longer the process will compute before completing its designated task.

9) How many and what type of resources the process has used

10) How many resources the process needs in order to complete

11) How many processes will need to be terminated

12) Whether the process is interactive or batch

Which term is best suited for the situation where several process access and manipulate shared data concurrently and final value of data depends which process finishes last.

Ans:

A situation like this, where several processes access and manipulate the same data concurrently and the outcome of the manipulation depends on the particular order in which the access takes place, is called a race condition.

Write the names of common file structures?

Ans:

A file has certain defined structure characteristics according to its type. A few common types of file structures are:

- Record Structures
- Complex structures

Source open software are helpful for testing algorithms as compared to pirated software?

Ans:

Open source software is very often developed in a public, collaborative manner. Open source software is the most prominent example of open source development and often compared to (technically defined) user-generated content or (legally defined) open content movements Reference: http://en.wikipedia.org/wiki/Open_source_software

Degree of multiprogramming when increased what effect is on CPU utilization?

Ans:

Multi-programming increases CPU utilization by organizing jobs so that the CPU always has one to execute. The operating system keeps several jobs in memory simultaneously.

Look algorithm?

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Ans:

This algorithm is a version of SCAN. In this algorithm the arm only goes as far as the last request in each direction, then reverses direction immediately, serving requests while going in the other direction. That is, it looks for a request before continuing to move in a given direction.

Disadvantage of dynamic loading?

Ans:

Disadvantage:

- However the run time activity involved in dynamic loading is a disadvantage.
- Dynamic programming does not require special support from the operating system.

What is mounting? Name two types of mounting. Give your answer with respect to File System?

Ans:

Mounting makes file systems, files, directories, devices, and special files available for use at a particular location. Mount point is the actual location from which the file system is mounted and accessed. You can mount a file or directory if you have access to the file or directory being mounted and write permission for the mount point.

These are the types of mounts:

- Remote mount
- Local mount

What do we name to an address that is generated by the CPU?

Ans:

A logical address is generated by CPU while a program is running. Since a logical address does not physically exists it is also known as a virtual address. This address is used as a reference by the CPU to access the actual physical memory location.

Which anamoly is invloed in FIFO page replacement?

Ans:

In computer storage, Bélády's anomaly is the phenomenon in which increasing the number of page frames results in an increase in the number of page faults for certain memory access patterns. This phenomenon is commonly experienced when using the first-in first-out (FIFO) page replacement algorithm.

The problem with using an acyclic-graph structure is ensuring that there are nocycles. What is the solution?

Ans:

One serious problem with using an acyclic-graph structure is ensuring that there are no cycles. A solution is to allow only links to files not subdirectories. Also every time a new link is added use a cycle detection algorithm to determine whether it is OK. If cycles are allowed, we want to avoid searching any component twice. A similar problem exists when we are trying to determine when a file can be deleted. A value of 0 in the reference count means no more references to the file/directory can be deleted. However, cycles can exist, e.g, due to self-referencing. In this case we need to use a garbage collection scheme, which involves traversing the entire file system, marking everything that can be accessed. Then a second pass collects everything that is not marked onto a

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list of free space. However this is extremely time consuming and is seldom used. However it is necessary because of possible cycles in a graph.

The main Purpose of the computer system is to run different programs, why we run these programs?

Ans:

The main purpose of a computer system is to execute programs. The programs, along with the data they access, must be in the main memory or primary storage during their execution. Since main memory is too small to accommodate all data and programs, and because the data it holds are lost when the power is lost, the computer system must provide secondary storage to backup main memory. Most programs are stored on a disk until loaded into the memory and then use disk as both the source and destination of their processing. Like all other resources in a computer system, proper management of disk storage is important.

Differ between physical address and virtual address?

Ans:

Real memory uses Physical addresses. These are the members that the memory chips react to on the bus. Virtual addresses are the logical addresses that refer to a process' address space. Thus, a machine with a 16-bit word can generate virtual addresses up to 64K, regardless of whether the machine has more or less memory than 64 KB.

How vfork system call differes from fork system call?

Ans:

In fork() system call, child and parent process have separate memory space. While in vfork() system call, child and parent process share same address space. 2. The child process and parent process gets executed simultaneously.

Briefly explain that how a page Fault occurs and who is responsible to handle page fault?

Ans:

A page fault occurs when a program attempts to access data or code that is in its address space, but is not currently located in the system RAM. So when page fault occurs then following sequence of events happens : The computer hardware traps to the kernel and program counter (PC) is saved on the stack. On Google

In case of file protection, what should the file owner/creator be able to control? And what operations need to be controlled?

Ans:

File owner/creator should be able to control

- What can be done
- By whom

Several types of operations may be controlled:

- **Read**: read from the file Write: write or rewrite to the file
- **Execute**: Load the file into memory and execute it
- **Append**: Write new information at the end of the file
- **Delete**: Delete the file and free its space for possible reuse
- **List**: List the name and attributes of the file.

Analyze that how an operating system protects the CPU?

Ans:

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CPU protection: CPU usage is protected by using the timer device, the associated timer interrupts, and OS code called the scheduler. While running in user mode, the CPU cannot change the timer value or turn off the timer interrupt, because these require privileged operations.

What is a File control block?

Ans:

A file control block is a memory data structure that contains most of the attributes of a file.

Under what conditions can you use the Wait-For graph to detect deadlock?

Ans:

If all resources have only a single instance, then we can define deadlock detection algorithm that uses a variant of the resource allocation graph, called a Wait-For graph.

Name any two schemes that allows implementation of page table?

Ans:

- In the CPU registers
- In the Main Memory

There are two dependent processes in Linux\ Unix operating system on same machine. how can these processes communicate among themselves?

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_____THE END_____